

Revinsight Assessment - Data Analysis

Q. Should We Hire a New Employee to Replace sp12?

Problem Statement

Data of a branded clothing store of 81 months is available. One of the sales staff, sp12, has just resigned. This document walks through the step-by-step logic to decide:

1. Should we hire a replacement for sp12?
2. How does the notice period temporary staff count spike affect the gross margin for that month?

Current standing business rules are:

- Always aim to have around 7 sales staff
 - Always try to maximize expected gross margin over the next 6 months
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The Gross Margin Formula

Monthly gross margin is calculated as:

$$\text{Gross Margin} = (50\% \times \text{Sales}) - (\text{Salary} \times \text{Staff Count}) - (5\% \times \text{Sales})$$

$$\text{Gross Margin} = 0.45 \times \text{Sales} - (3 \times \text{Staff Count})$$

Over a 6-month period:

$$\text{Total Gross Margin} = 0.45 \times 6 \times \text{avg_monthly_Sales} - 18 \times \text{Staff_Count}$$

The avg_monthly_sales is average of monthly sales for a give staff_count

Why We Don't Judge by sp12's Individual Performance

The intuitive first instinct is to look at sp12's personal sales history — if she was a high performer, find someone to cover her sales gap; if she was below average, don't bother. This logic fails for one simple reason: a new hire's future performance is completely unknown. There is no way to guarantee a replacement will match sp12's numbers. The decision must therefore be based on what the store earns at different staffing levels, not on the individual leaving.

The Approach

Instead of individual performance, we use historical store-level data grouped by staff count to find the empirical average monthly sales at each headcount. This captures real-world effects like floor crowding, customer service capacity, and diminishing returns automatically.

From the dataset (81 months of history):

Staff Count	Avg Monthly Sales
3	23.10
5	41.59
7	104.59
8	110.00

Question 1: Should We Hire a Replacement?

When sp12 leaves, we will have 6 staff. The question is whether adding 1 person back — going from 6 to 7 — improves our gross margin over the next 6 months.

We compare two flat scenarios across all 6 months:

Scenario	Staff	Monthly Sales
Hire	7	avg_sales at 7
Don't Hire	6	avg_sales at 6

The change in gross margin from hiring is:

$$\Delta \text{ Gross Margin} = 6 \times [0.45 \times (\text{sales_at_7} - \text{sales_at_6}) - 3]$$

Let $X = \text{sales_at_7} - \text{sales_at_6}$ (the incremental monthly sales one extra person brings).

$$\Delta \text{ Gross Margin} = 0.45 \times 6 \times X - 18$$

Decision rule:

- If $0.45 \times 6 \times X > 18 \rightarrow$ Hire (the extra sales outweigh the salary cost)
- If $0.45 \times 6 \times X < 18 \rightarrow$ Do not hire (salary costs more than the sales gained)

The breakeven point for X is:

$$X = 18 / (0.45 \times 6) = 6.67 \text{ units/month}$$

If the 7th person adds more than 6.67 units(derived from historical data) of monthly sales on average, hiring is profitable.

Question 2: The effect of the Notice period model?

Assuming Question 1 says hire(where we don't properly account for the staff_count in notice period in real world), we now ask: how does the gross margin get affected in the notice period month due to presence of the resigned and new hire simultaneously?

Our current practice ("overlap model") has the new hire join while sp12 is still working. This means for one month the store runs with one extra person compared to steady state.

The two models only differ in Month 1. Months 2–6 are identical — both models have the same staff count after sp12 leaves — so they cancel out of the comparison. Only the notice month matters.

Month	Overlap Model	No-Overlap Model
Month 1	sc + 1 staff	sc staff
Months 2–6	sc staff (identical)	sc staff (identical)

Where sc = steady-state headcount before the resignation (e.g., 7).

The delta for the notice month only:

$$\Delta \text{Overlap} = 0.45 \times (\text{sales_at_sc}+1 - \text{sales_at_sc}) - 3$$

Using the data:

$$\text{sales_at_8} - \text{sales_at_7} = 110.00 - 104.59 = 5.41 \text{ units/month}$$

$$\Delta \text{Overlap} = 0.45 \times 5.41 - 3 = 2.43 - 3 = -0.57 \text{ units}$$

The overlap month(notice period) produces a net margin loss of 0.57 units. The 8th person during the notice period generates less incremental sales than their monthly salary costs.

Notice month gross margin effect: The overlap model slightly hurts margin during the notice month.

Recommendations

On replacing sp12: Run the `get_recommendation()` function with the actual dataset to get the precise delta. The framework above gives the decision rule — whether the 7th person sales has +ve effect on the gross margin, i.e hire or not.

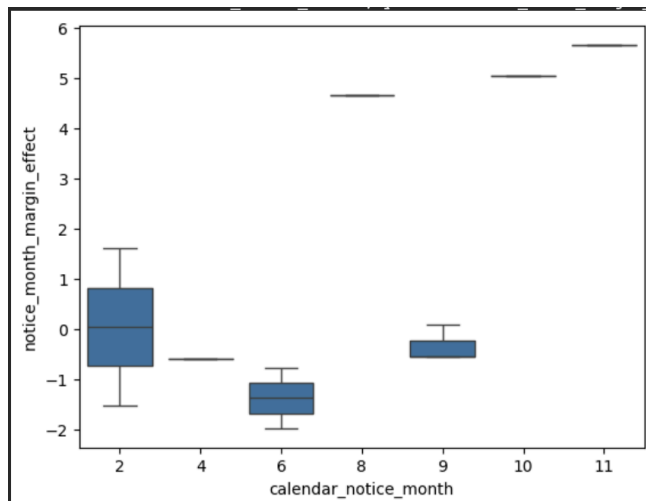
On the overlap model: Based on the historical average sales data, having 8 staff during the notice month of sp12 decreases gross margin by approximately 0.57 units compared to waiting. Two alternatives to consider:

1. Have the new hire start immediately after sp12's last day rather than during the notice period.
2. If some onboarding overlap is operationally valuable (training, handover), bring the new hire in as a part-time employee during the notice month at a reduced salary, so the gross margin increases compared to 8 full timers (gross margin= $0.45 \times \text{sales} - (7 \times 3) - 1 \times (\text{part-time salary})$)

NOTE: Below you can see that having the `notice_month_margin_effect` are +ve(i.e having 8th employee also lead to increase in gross margin) for certain months(seasonal patterns), which could be months being peak seasons for sales. Sometimes the quantity is low sometimes high.

Sales member	active_period	notice_month	notice_month_margin_effect	Calendar_notice_month = ((notice_month-1)%12)+ 1; (For observing seasonal patterns)
sale sp1	66	66	-1.983478	6
sale sp2	30	30	-0.782364	6
sale sp3	46	46	5.037831	10
sale sp4	34	35	5.660232	11
sale sp5	19	20	4.665499	8
sale sp6	36	38	1.602425	2
sale sp7	55	57	0.071990	9
sale sp8	62	81	-0.562794	9
sale sp9	47	76	-0.592105	4
sale sp10	47	81	-0.562794	9
sale sp11	25	62	-1.524139	2

sale sp12	36	81	-0.562794	9
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Limitations of the Current Approach

The Problem: The current model mixes the entire year together. This flattens out seasonal peaks, **capping the true sales potential** and making busy holiday rushes look identical to slow summers.

The Solution: Switch to a **Seasonal Approach** by calculating historical averages using only the n months directly ahead (e.g., if evaluating in October, use only past October–March data).

Why It Matters: This transforms the tool into a **staffing planner** that reveals if a festival rush justifies extra help, letting you calculate *exactly how many* seasonal workers are profitable to bring on.

The Data Catch: The current 81-month dataset only provides 11–12 examples per individual calendar month. While fine for a yearly average, we need a larger dataset to make these seasonal filters highly accurate.